PHD125 Pharmaceutical Organic Chemistry · Carboxylic Acid, Ester, Amines, Amide

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

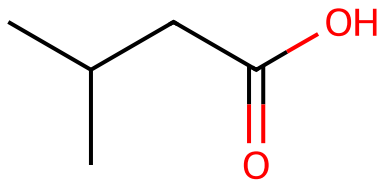
About 30 minutes. Questions follow the four subchapters of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. The answer key is at the back.

4.1 · Carboxylic acids

Q1 CH4-LO1

Name this acid.



A. 3-methylbutanoic acid

B. 2-methylbutanoic acid

C. 3-methylpropanoic acid

D. pentanoic acid

Q2 CH4-LO1

Match each common name to its IUPAC name.

Options: methanoic acid · ethanoic acid · propanoic acid · butanoic acid · pentanoic acid

formic acid _____

acetic acid _____

propionic acid _____

butyric acid _____

Q3 CH4-LO2

Which is the strongest acid? (pK_a in brackets)

A. chloroacetic acid (2.85)
B. acetic acid (4.74)
C. benzoic acid (4.19)
D. formic acid (3.74)

Q4 CH4-LO2

Name the salt formed when propanoic acid reacts with sodium hydroxide.

_____

Q5 CH4-LO2

Acetic acid reacts with sodium bicarbonate to release carbon dioxide gas.

True / False

4.2 · Esters

Q6 CH4-LO3

Name $CH_3COOCH_2CH_2CH_3$.

A. propyl ethanoate (propyl acetate)
B. ethyl propanoate
C. methyl butanoate
D. propyl methanoate

Q7 CH4-LO3

Which acid and alcohol make methyl butanoate (apple smell)?

A. butanoic acid + methanol
B. methanoic acid + 1-butanol
C. butanoic acid + ethanol
D. propanoic acid + methanol

Q8 CH4-LO3

Give the products of ethyl acetate under each condition.

Options: acetic acid + ethanol · sodium acetate + ethanol · acetamide + ethanol

water + acid catalyst, heat _____

NaOH solution, heat _____

Q9 CH4-LO3

Esters boil higher than carboxylic acids of the same molar mass.

True / False

4.3 · Amines

Q10 CH4-LO4

Classify each amine.

Options: 1° · 2° · 3°

$CH_3CH_2NH_2$ _____

$(CH_3)_2NH$ _____

$(CH_3)_3N$ _____

$(CH_3)_2CHNH_2$ _____

Q11 CH4-LO4

Name $CH_3CH_2NHCH_2CH_3$.

_____

Q12 CH4-LO4

Which has the highest boiling point? (all about 73–74 g/mol)

A. butyl alcohol
B. butylamine
C. diethylamine
D. diethyl ether

Q13 CH4-LO4

Many amine drugs are sold as hydrochlorides because the salt:

A. is ionic and water-soluble
B. is a stronger drug
C. is an ester
D. evaporates more easily

Q14 CH4-LO4

Alkaloids are extracted from plants with dilute acid because:

A. they are bases that form water-soluble salts
B. they are acids
C. acid destroys the plant
D. they are esters

4.4 · Amides

Q15 CH4-LO5

Name $CH_3CH_2CONH_2$.

A. propanamide
B. propylamine
C. propanoic acid
D. ethanamide

Q16 CH4-LO5

Acetic acid + methylamine, heated, gives:

A. N-methylacetamide + water
B. methyl acetate + ammonia
C. methylammonium acetate only, never an amide
D. acetamide + methanol

Q17 CH4-LO5

Amides are bases, like amines.

True / False

Q18 CH4-LO5

Give the products of hydrolysing butyramide under each condition.

Options: butyric acid + NH_4^+ · butyrate ion + NH_3 · butanol + NH_3

acid (H_3O^+), heat _____

base (OH^-), heat _____

Pharmacy scenario · The medicine cabinet

Scenario

Puan Salmah brings in a bag from her late father's medicine cabinet: an old bottle of **aspirin** that smells of vinegar, a tube of **lidocaine hydrochloride** gel, and some **effervescent** vitamin C tablets that fizz in water. She asks which ones are still safe to use.

Q19 CH4-LO3

What has happened to the aspirin?

A. Its ester group has hydrolysed to salicylic acid and acetic acid

B. It has oxidised to an aldehyde

C. It has turned into an amide

D. Nothing; the smell is normal

Q20 CH4-LO2

The tablets fizz in water because:

A. an acid reacts with sodium bicarbonate to release CO_2

B. an ester is hydrolysed

C. an amine is protonated

D. the water boils

Q21 CH4-LO4

Lidocaine is supplied as the hydrochloride because the salt forms at its:

A. amine nitrogen
B. amide C=O
C. benzene ring
D. methyl groups

Q22 CH4-LO3

What would you advise Puan Salmah?

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PHD125 Pharmaceutical Organic Chemistry · Carboxylic Acid, Ester, Amines, Amide

After class: check your understanding

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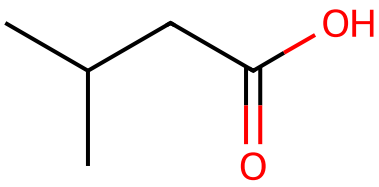
About 30 minutes. Questions follow the four subchapters of the lecture and end with a pharmacy scenario.

Answer key

4.1 · Carboxylic acids

Q1 CH4-LO1

Name this acid.



A. 3-methylbutanoic acid

B. 2-methylbutanoic acid

C. 3-methylpropanoic acid

D. pentanoic acid

Four-carbon chain including the carboxyl carbon (C1); the methyl is on C3.

Q2 CH4-LO1

Match each common name to its IUPAC name.

formic acid → methanoic acid; acetic acid → ethanoic acid; propionic acid → propanoic acid; butyric acid → butanoic acid

Pentanoic acid is valeric acid (Ball 15.1).

Q3 CH4-LO2

Which is the strongest acid? (pK_a in brackets)

A. chloroacetic acid (2.85)

B. acetic acid (4.74)

C. benzoic acid (4.19)

D. formic acid (3.74)

Smallest pK_a = strongest acid (FGOB 17.2).

Q4 CH4-LO2

Name the salt formed when propanoic acid reacts with sodium hydroxide.

sodium propanoate

Cation first; -ic acid → -ate.

Q5 CH4-LO2

Acetic acid reacts with sodium bicarbonate to release carbon dioxide gas.

True / False → **True**

$RCOOH + NaHCO_3 \rightarrow RCOO^-Na^+ + H_2O + CO_2$.

4.2 · Esters

Q6 CH4-LO3

Name $CH_3COOCH_2CH_2CH_3$.

A. propyl ethanoate (propyl acetate)

B. ethyl propanoate

C. methyl butanoate

D. propyl methanoate

Propyl on the O; acetate (2 C) acid part.

Q7 CH4-LO3

Which acid and alcohol make methyl butanoate (apple smell)?

A. butanoic acid + methanol

B. methanoic acid + 1-butanol

C. butanoic acid + ethanol

D. propanoic acid + methanol

The alkyl on the O comes from the alcohol.

Q8 CH4-LO3

Give the products of ethyl acetate under each condition.

water + acid catalyst, heat → acetic acid + ethanol; NaOH solution, heat → sodium acetate + ethanol

Acidic hydrolysis gives the acid; basic hydrolysis (saponification) gives the salt.

Q9 CH4-LO3

Esters boil higher than carboxylic acids of the same molar mass.

True / False → **False**

Esters cannot hydrogen bond to each other; methyl acetate 57 °C vs propanoic acid 141 °C.

4.3 · Amines

Q10 CH4-LO4

Classify each amine.

$CH_3CH_2NH_2$ → 1°; $(CH_3)_2NH$ → 2°; $(CH_3)_3N$ → 3°; $(CH_3)_2CHNH_2$ → 1°

Count the carbon groups on the nitrogen.

Q11 CH4-LO4

Name $CH_3CH_2NHCH_2CH_3$.

diethylamine

Two ethyl groups on N (FGOB 16.2).

Q12 CH4-LO4

Which has the highest boiling point? (all about 73–74 g/mol)

A. butyl alcohol

B. butylamine

C. diethylamine

D. diethyl ether

O–H hydrogen bonds are stronger than N–H (FGOB 16.3).

Q13 CH4-LO4

Many amine drugs are sold as hydrochlorides because the salt:

A. is ionic and water-soluble

B. is a stronger drug

C. is an ester

D. evaporates more easily

Amine salts dissolve in water and are stable, odourless solids.

Q14 CH4-LO4

Alkaloids are extracted from plants with dilute acid because:

A. they are bases that form water-soluble salts

B. they are acids

C. acid destroys the plant

D. they are esters

Adding base afterwards frees the alkaloid again (FGOB 16.7).

4.4 · Amides

Q15 CH4-LO5

Name $CH_3CH_2CONH_2$.

A. propanamide

B. propylamine

C. propanoic acid

D. ethanamide

-oic acid → -amide.

Q16 CH4-LO5

Acetic acid + methylamine, heated, gives:

A. N-methylacetamide + water

B. methyl acetate + ammonia

C. methylammonium acetate only, never an amide

D. acetamide + methanol

The –OH of the acid and an H on N are lost as water (FGOB 17.3).

Q17 CH4-LO5

Amides are bases, like amines.

True / False → **False**

The N lone pair is drawn towards the C=O, so amide solutions are neutral (Ball 15.14).

Q18 CH4-LO5

Give the products of hydrolysing butyramide under each condition.

acid (H_3O^+), heat → butyric acid + NH_4^+; base (OH^-), heat → butyrate ion + NH_3

In acid the ammonia is protonated; in base the acid is deprotonated (FGOB 17.4).

Pharmacy scenario · The medicine cabinet

Q19 CH4-LO3

What has happened to the aspirin?

A. Its ester group has hydrolysed to salicylic acid and acetic acid

B. It has oxidised to an aldehyde

C. It has turned into an amide

D. Nothing; the smell is normal

Moisture hydrolyses the ester; acetic acid is the vinegar smell.

Q20 CH4-LO2

The tablets fizz in water because:

A. an acid reacts with sodium bicarbonate to release CO_2

B. an ester is hydrolysed

C. an amine is protonated

D. the water boils

$RCOOH + NaHCO_3 \rightarrow RCOO^-Na^+ + H_2O + CO_2$.

Q21 CH4-LO4

Lidocaine is supplied as the hydrochloride because the salt forms at its:

A. amine nitrogen

B. amide C=O

C. benzene ring

D. methyl groups

The 3° amine accepts H^+; the amide nitrogen is not basic.

Q22 CH4-LO3

What would you advise Puan Salmah?

Do not use the aspirin: it has degraded by hydrolysis and may also be past its expiry date; return it to the pharmacy for safe disposal. Check the expiry dates of the lidocaine gel and the effervescent tablets and use them only if in date and stored properly. Store medicines in a cool, dry place, not a humid bathroom.

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